Introduction :
This script was developped to ease the process of creating packaging mock-up with laser engraver from a diecut designed with Adobe Illustrator. It works paired with the template Corius_SVG_GCODE_for_LASERGRBLv2.ait

The script addresses the issue of Illustrator natively exported SVG files being not always compatible with LaserGRBL. The SVG generated with this script are only converting path stroke (no text, no fill, no stroke effect like dash, no color accuracy) with the only purpose to import them into LaserGRBL (they may work with other GRBL software though).
It also exports into GCODE files, with some GRBL settings directly changeable within Illustrator such as laser power, feed speed and number of pass.

The scipt was developped in Extendsrcript, with Adobe ExtendScript Toolkit CC.

Usage principle :
The script will use the paths from the main layers, separating different laser job settings, to generate SVG files as well as GCODE files (.nc file extension) which can both be loaded into LaserGRBL software. The main GCODE parameters can be changed directly from the Illustrator template on the “GCODE_PARAMS” layer.

Changelog in v2 :

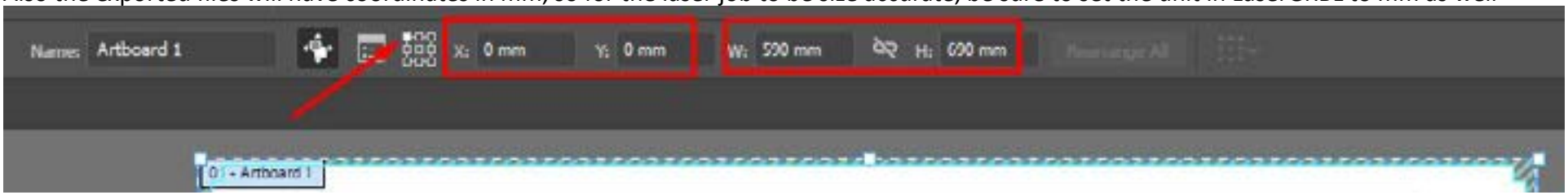
Category	Function	Description
Main layers	Creation of laser job	• The number of main layers is now unlimited. • Their names aren’t predetermined, you can use whatever names you want for these layers as long as the names are matched by the job names in the “GCODE_PARAMS” layer. Any layer with a name not listed in the Material Laser Job Names will simply be ignored.
Artboard	Creation of laser job	• The script now considers only the paths that are fully within the Artboard limits, it will ignore the paths partially or fully outside, to ensure the SVG/GCODE won’t send laser job coordinates outside of your laser engraver working area.
File export	SVG export settings	• You can’t export merged SVG file anymore. • You can now choose to export or not the SVG files
File export	GCODE export settings	• You can now choose to export or not the gcode files • You can choose to export both the individual and the merged gcode files in one run of the script • For the merge gcode files you can choose if the laser jobs are kept separated or mixed to better optimize the time

Category	Function	Description
Preview	Time estimate	• You can have a preview of the Laser movements within Illustrator with a time estimate for the laser job to complete in LaserGRBL
Material profile	Laser Job Order	• You can set the order in which the jobs will be done in merged files
Material profile	Preview	• You can activate or deactivate the preview of the job
Material profile	Convert job to dashed path	• You can convert each job independently from a continuous path to a dashed path in gcode and svg, while overwriting the laser power and speed to use different settings in dashed than in continuous
Material profile	Merge job	• You can specify for each job if it can be merged with other jobs when export merge file is selected, or if this particular job needs to be kept unmerged even when merging files.
Paths processing	Laser head travel optimization	• The whole SVG/GCODE conversion has been redesigned to eliminate most of the unnecessary laser head travel, which should reduce a bit the engraving time compared to v1

Template configuration :

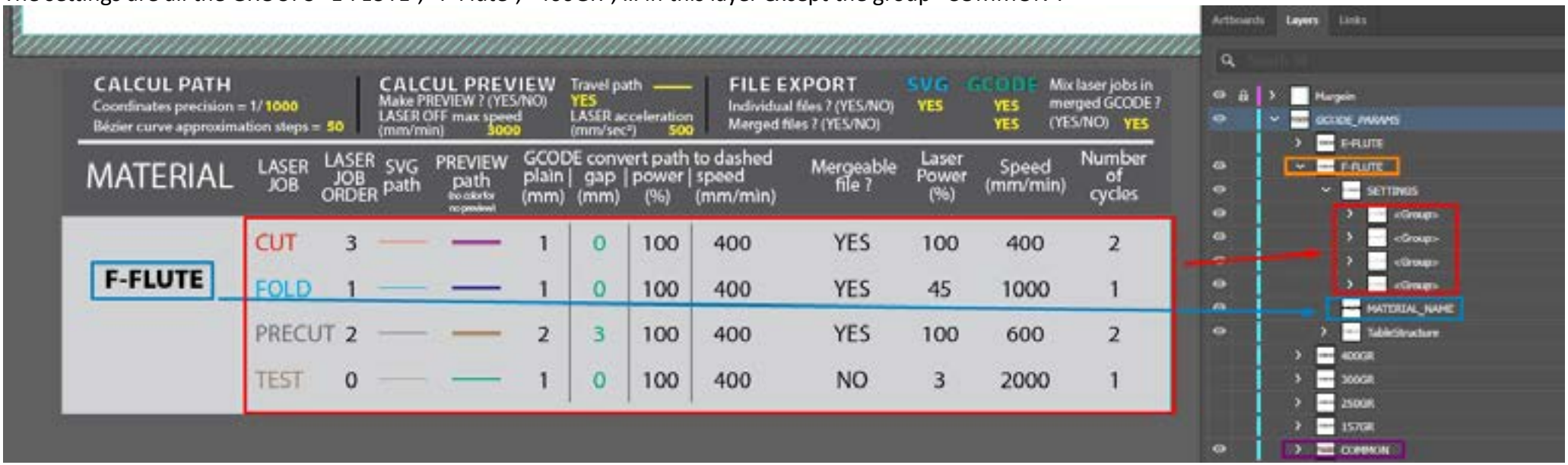
The first thing to do is to configure the template accordingly to your laser engraver, and save it back into a template file (.ait) so you don't have to configure it again each time you need to create a new diecut mock-up.

The Artboard : make sure your artboard is set to the maximum engraving width/heigth of your engraver, and that the top left corner is at the x,y position 0,0 in Illustrator. The script implements a unit conversion into mm, but it was not fully tested and might not work properly, so it is safer to use the script with the Illustrator file unit set to mm.
Also the exported files will have coordinates in mm, so for the laser job to be size accurate, be sure to set the unit in LaserGRBL to mm as well



The margein : if your laser engraver is like mine, i.e a D.I.Y. kind with poor cable management you may need a security margein on the side to prevent some issue like cable pinched between the engraver frame and the motor, preventing the laser head to go fully back to the 0,0 position, resulting in a botched laser cut. In this case, prepare the "Margein" layer to make obvious the usable area. The script will simply ignore this layer, so don't use it to prepare your diecut. When your "Margein" layer is finished, lock it so you won't mess it up afterward.

The “GCODE_PARAMS” layer : I already prepared some material profiles in this layer. You will need to adapt them and/or create your own profiles here. The settings are all the GROUPS “E-FLUTE”, “F-Flute”, “400GR”, … in this layer except the group “COMMON”.



Here is how it works :
In this layer the group “COMMON” must remain named exactly as is, and all the items in it must remain with their own names. The other groups can all be renamed or deleted except at least one. They are the material profiles.
Name the groups of material profiles accordingly to the materials you will use.
All the material profile groups must keep the same structure. Example with the group “F-FLUTE” :
- the group name is not checked by the script. Instead, the script will check the named textfield “MATERIAL_NAME” content (the blue frame on the left “F-FLUTE” in the example) to retrieve the material name.

- the subgroup “TableStructure” is ignored by the script. Its purpose is only to present the things nicely for the user in Illustrator.
- the subgoup “SETTINGS” contains 1 subgroup for each laser job. In my example there is 4 laser jobs (“CUT”, “FOLD”, “PRECUT”, “TEST”), and each has its own group. If you need more laser jobs, simply duplicate 1 group and place it below.

Explanation of each setting in the laser job groups :
- LASER JOB : type in the name of the kind of engraving you want. This name will be added in the exported file name to differenciate each laser job files.
- LASER JOB ORDER : The order in which the laser jobs will be run in merged files. They will run from the lowest to the highest number. Numbers don’t need to be consecutive, and don’t need to start at 0. This setting is ignored in individual file export.
- SVG PATH : the color of this stroke determine the color to use for this laser job path in the SVG eported file. Note : the template is in CMYK color mode, and SVG files are in RGB. The color conversion from the Illustrator file to the SVG with this script is very basic and far from accurate.
- PREVIEW PATH : the color and thickness of this stroke determine the color and thickness to use for the preview path of this laser job.
- GCODE convert path to dashed - **plain** : length of the dash in mm
- GCODE convert path to dashed - **gap** : length of the gap in mm. **Write a gap value of 0 to prevent the path from being converted to dashed path. Any number bigger than 0 will trigger the path conversion to dash automatically**
- GCODE convert path to dashed - **power** : power of the laser to use for the dashed converted path in replacement of the Laser Power setting
- GCODE convert path to dashed - **speed** : speed of the laser to use for the dashed converted path in replacement of the Speed setting
- Mergeable file ? : Write YES to allow this Laser Job to be merged with others in exported file, or NO to ensure this job will be exported in a standalone file.
- Laser Power (%) : Write the power in % of the laser to use for this job
- Speed (mm/min) : Write the movement speed of the laser head to use for this job
- Number of cycles : Write the number of time this job needs to be repeated. **Note : different laser job can’t be mixed in merged file if their “Number of cycles” is different.**

Explanation of the settings in the “COMMON” group :
- CALCUL PATH:

Coordinates precision : by default 1/1000th, so x,y coordinates in SVG and GCODE will be rounded to the 1/1000th. You might need to change this to 1/100th or 1/10th if your laser engraver can’t take as precise inputs.

Bézier curve approximation steps : to maximize compatibility the curves are converted into multiple straight segments. If you have very long curves you might want to increase the steps count, otherwise 50 is already plenty.

- CALCUL PREVIEW:

Make PREVIEW ? (YES/NO) : Write YES to create a layer to preview your laser jobs for each GCODE export, or NO if you don’t need a preview within Illustrator. The preview layer will be created below all the other layers.

LASER OFF max speed (mm/min) : Write the fast travel speed as it is set in LaserGRBL, this will be used to calculate a time rough estimate for the laser job to complete.

LASER acceleration (mm/sec²) : Write the acceleration ratio as it is set in LaserGRBL, this will help to give a better time estimate for the laser job to complete.

Travel path : the color and thickness of this stroke determine the color and thickness to use for the preview path of Laser OFF movement

Note : You can't have a preview if you don't have at least 1 GCODE file exported.

Note : If the "Make PREVIEW" is set to NO, there won't be a time estimate.

- FILE EXPORT:

Individual files ? (YES/NO) - SVG : Write YES to export each Laser Job in indiviual SVG file, and NO to not export SVG files

Individual files ? (YES/NO) - GCODE : Write YES to export each Laser Job in indiviual GCODE file, and NO to skip the export of individual GCODE files

Merged files ? (YES/NO) - GCODE : Write YES to export Laser Job merged GCODE file, and NO to skip the export of individual GCODE files

Mix laser jobs in merged GCODE ? (YES/NO) : Write YES to mix the Laser Job in merged GCODE file for better time optimization, and NO to keep each laser job separated in merged GCODE files. This setting is ignored if "Merged files ?" setting is set to NO.

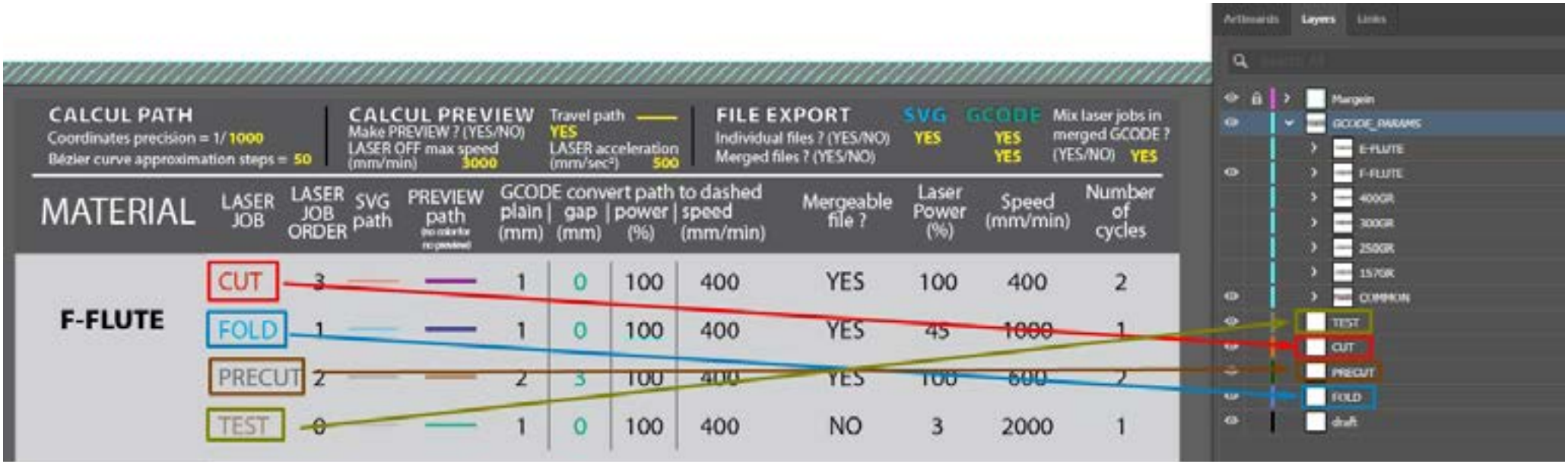
Note : different laser jobs won't be mixed if their "Number of cycles" is different.

Once you're done with the settings, save your template as a new template so you don't have to re-set it each time.

Diecut creation :

you can use the layer "draft" to create any shape/size/guide to help you achieve the proper diecut, and you can add as many draft layers as you need, with or without naming them, they will be disregarded by the script.

For the actual laser paths you want to export, be sure to draw the paths on the corresponding and properly named layer, i.e using the exact same name as the "LASER JOB" name in the Material profile, like in below example.



When the whole diecut is complete, in order to create the corresponding SVG and GCODE, you need to “select” the material preset to use by simply unhiding the corresponding group in the “GCODE_PARAMS” layer and hiding the others : you can only run the export for one material profile at a time.

Check the CALCUL PREVIEW > Make PREVIEW setting, as well as the whole FILE EXPORT settings, then run the script.

Run the script :

The script file should be copy/pasted into the folder : C:\Program Files\Adobe\Adobe Illustrator 2026\Presets\en_US\Scripts
(Change the red text according to your Illustrator version and language), then you can run it by going into Illustrator menu :
File > Scripts > Diecut_To_SVGandGCode-V2

The script will export SVG files and GCODE files in subfolders “SVG\” and “GCODE\” of your illustrator file.
The file names will be : [the ai file name]_[material]_[LaserJob(s)]exportType, with exportType being “(indiv)” or “(merged)” or “(mixed)”.
Example :
The Illustrator file name is HeadsetColorBox.ai
The selected Material is F-FLUTE
The Laser Jobs are FOLD, PRECUT, and CUT
The export type is merged
The resulting gcode file name is : HeadsetColorBox_F-FLUTE_FOLD+PRECUT+CUT(merged).nc
Let’s say the FOLD laser job is set with parameter “Mergeable” = NO, and both CUT and PRECUT are set to YES. Then you’ll have 2 files :
- HeadsetColorBox_F-FLUTE_FOLD(merged).nc
- HeadsetColorBox_F-FLUTE_PRECUT+CUT(merged).nc

If your material profile has laser jobs set with “Number of cycles” > 1 then the GCODE will baked-in as many run as this number for the corresponding laser jobs.

The script will pop-up a window to list all the files it has exported.

Finally, if the setting “Make PREVIEW ?” is set to YES, the laser head movements will be shown, each with the color corresponding to the laser job, on a dedicated layer for each ExportType :
- Individual : if Individual files ? (YES/NO) - GCODE is set to YES
- Merged : if Merged files ? (YES/NO) - GCODE is set to YES and Mix laser jobs in merged GCODE ? (YES/NO) is set to NO
- Mixed : if Merged files ? (YES/NO) - GCODE is set to YES and Mix laser jobs in merged GCODE ? (YES/NO) is set to YES

Each preview layer will also display a rough time estimate.

Copyright and stuff :

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